



WILDCARD

ML DOMAIN · RECRUITMENT ROUND

“Decode the data. Earn your place.”

DATASETS	3 Datasets provided — Bank Marketing · Telco Customer Churn · Student Performance Index
DOMAIN	Banking & Finance Telecom Education Analytics — ML Domain
TASK TYPE	Binary Classification (Bank Marketing, Telco Churn) & Regression (Student Performance Index)
ROWS	11,163 7,044 10,000 rows respectively (see Section 2)
SUBMIT VIA	GitHub Repository Link (Google Form)



1 Task Overview

This is the MLCOE ML Domain Wildcard Recruitment Round. You will be given a real-world dataset and must build a complete Machine Learning pipeline from scratch — from data exploration to a saved, evaluated model.

Datasets	3 datasets provided (Modified by MLCOE) — see Section 2 for full details
Task	Predict the target column for your assigned dataset (classification or regression)
Problem Type	Binary Classification (Bank Marketing / Telco Churn) or Regression (Student Performance Index)
Language	Python only
Environment	Jupyter Notebook (.ipynb) — Google Colab recommended
Submission	GitHub Repository link submitted via Google Form

Note: This round now includes THREE datasets across two problem types. Pick ONE dataset according to your own choice and proceed further. Do not mix datasets or use external/alternative versions of any of them.

Dataset Download Links

- Bank Marketing Dataset —
<https://drive.google.com/file/d/13QODZc4Gmnts6i2UNzrzi815XdyZZaPA/view?usp=sharing>
- Telco Customer Churn Dataset —
https://drive.google.com/file/d/15y0luljqP_rdx6WyW-eBWP0sxu38poYs/view?usp=sharing
- Student Performance Index Dataset —
https://drive.google.com/drive/folders/1FhJYMFicSXau_C7icWsD9XRT8TrwT-iX?usp=drive_link



2A About the Dataset — Bank Marketing

CLASSIFICATION · Bank Marketing Dataset

11,163 rows | 17 columns | Target: **deposit**

The dataset contains data from direct marketing campaigns (phone calls) of a banking institution. The goal is to predict if a client will subscribe to a term deposit — stored in the column **deposit**.

Column	Type	Description
age	Numerical	Age of the client
job	Categorical	Type of job
marital	Categorical	Marital status
education	Categorical	Education level
default	Binary	Has credit in default?
balance	Numerical	Average yearly bank balance (euros)
housing	Binary	Has housing loan?
loan	Binary	Has personal loan?
contact	Categorical	Contact communication type
day	Numerical	Last contact day of the month
month	Categorical	Last contact month of year
duration	Numerical	Last contact duration in seconds
campaign	Numerical	Number of contacts during this campaign
pdays	Numerical	Days since client was last contacted (-1 means never)
previous	Numerical	Number of contacts before this campaign
poutcome	Categorical	Outcome of previous marketing campaign
deposit — TARGET	Binary	Has the client subscribed to a term deposit? (yes / no)

Note: Check the class distribution of 'deposit' in your EDA — handle any imbalance appropriately.

2B About the Dataset — Telco Customer Churn

CLASSIFICATION · Telco Customer Churn Dataset

7,044 rows | 21 columns | Target: **Churn**

The dataset contains customer account information for a telecom provider. The goal is to predict whether a customer will churn (leave the service) — stored in the column **Churn**.

Column	Type	Description
customerID	Identifier	Unique customer ID (drop before modelling)
gender	Categorical	Customer gender
SeniorCitizen	Binary	Whether the customer is a senior citizen
Partner	Binary	Whether the customer has a partner
Dependents	Binary	Whether the customer has dependents
tenure	Numerical	Number of months the customer has stayed
PhoneService	Binary	Whether the customer has phone service
MultipleLines	Categorical	Whether the customer has multiple lines
InternetService	Categorical	Customer's internet service provider type
OnlineSecurity	Categorical	Whether the customer has online security add-on
OnlineBackup	Categorical	Whether the customer has online backup add-on
DeviceProtection	Categorical	Whether the customer has device protection add-on
TechSupport	Categorical	Whether the customer has tech support add-on
StreamingTV	Categorical	Whether the customer has streaming TV
StreamingMovies	Categorical	Whether the customer has streaming movies
Contract	Categorical	Contract term of the customer
PaperlessBilling	Binary	Whether the customer has paperless billing
PaymentMethod	Categorical	Customer's payment method
MonthlyCharges	Numerical	Amount charged to the customer monthly
TotalCharges	Numerical	Total amount charged to the customer
Churn — TARGET	Binary	Whether the customer churned (Yes / No)

Note: 'TotalCharges' may load as text with blank entries — convert to numeric and handle missing values. Check the class distribution of 'Churn' in your EDA.



2C About the Dataset — Student Performance Index

REGRESSION • Student Performance Index Dataset

10,000 rows | 6 columns | Target: **Performance_Index**

The dataset contains records of student study habits and exam preparation. The goal is to predict a student's overall performance index — this is a Regression problem, not classification.

Column	Type	Description
Hours_Studied	Numerical	Number of hours the student studied
Previous_Scores	Numerical	Score obtained in previous assessments
Extracurricular_Activities	Categorical	Whether the student participates in extracurricular activities (Yes / No)
Sleep_Hours	Numerical	Average number of hours the student sleeps
Sample_Question_Papers_Practiced	Numerical	Number of sample question papers practiced
Performance_Index — TARGET	Numerical	Overall performance index of the student (target for regression)

Note: This dataset requires a Regression approach — class balance does not apply. Instead check the distribution/skewness of 'Performance_Index' and consider a log-transform if heavily skewed.



3 Mandatory Libraries

You must import and use only the following libraries. No other ML libraries are allowed unless listed in the Bonus section.

MUST COMPLETE

Library	Purpose	Import Statement
numpy	Numerical computations	import numpy as np
pandas	Data manipulation	import pandas as pd
matplotlib	Basic plotting	import matplotlib.pyplot as plt
seaborn	Statistical visualisation	import seaborn as sns
scikit-learn	ML models, metrics, preprocessing	from sklearn... (as needed)
joblib	Save and load trained model	import joblib

Note: First code cell of your notebook must import ALL of the above libraries together. Scattered imports will be penalised.

4 Task Instructions

Complete the following tasks in order inside a single Jupyter Notebook, using your assigned dataset only. Each task must be preceded by a clearly labelled Markdown cell.

Task 1 — Import Libraries & Load Dataset

MUST COMPLETE

- Import all mandatory libraries in the very first code cell.
- Load your assigned dataset, e.g. `df = pd.read_csv('bank.csv') / 'telco_churn.csv' / 'student_performance.csv'`
- Print `df.shape`, `df.head()`, `df.info()`, and `df.describe()`.

Task 2 — Exploratory Data Analysis (EDA)

MUST COMPLETE

- Check and report missing values — count and percentage per column.
- Check for duplicate rows and remove if found.
- Plot distribution of all numerical columns (histogram + KDE).



- Plot value counts for all categorical columns (bar chart).
- Plot the target column's distribution — for Bank Marketing / Telco Churn show class balance (deposit / Churn); for Student Performance Index show the distribution/skew of 'Performance_Index'.
- Plot a correlation heatmap for numerical features.
- Identify and report outliers using IQR method and box plots.

Task 3 — Feature Engineering

MUST COMPLETE

- Encode all categorical columns — use Label Encoding or One-Hot Encoding. Justify your choice.
- Scale numerical features using StandardScaler or MinMaxScaler.
- Split the dataset: 80% train / 20% test with random_state=42. Use stratify=y for the classification datasets (Bank Marketing / Telco Churn); stratify does not apply for the Student Performance Index regression dataset.

Task 4 — Model Training

MUST COMPLETE

Train models appropriate to your assigned dataset's problem type. The specific algorithm within each category is your choice — research and justify why you picked it.

Category	Classification (Bank Marketing / Telco Churn)	Regression (Student Performance Index)
Linear model	Logistic Regression	Linear / Ridge / Lasso Regression
Tree-based model	Decision Tree Classifier	Decision Tree Regressor
Ensemble model	Random Forest Classifier	Random Forest Regressor

Note: You only need to complete the column matching your assigned dataset's problem type — at least ONE model from each of the three categories.



Task 5 — Model Evaluation

MUST COMPLETE

Evaluate every model using the metrics appropriate to your problem type:

Metric Set	Details
Classification (Bank Marketing / Telco Churn)	Accuracy, Precision / Recall, and Confusion Matrix (plotted as heatmap, showing TP, TN, FP, FN for every model). Note accuracy alone can be misleading on imbalanced data.
Regression (Student Performance Index)	MAE (Mean Absolute Error), RMSE (Root Mean Squared Error), and R^2 Score for every model. Also plot Actual vs Predicted values / a residual plot.

Task 6 — Save Model with Joblib

MUST COMPLETE

- State clearly which model you chose as best and WHY based on the metrics.
- Save your best model using joblib:

```
import joblib
joblib.dump(best_model, 'best_model.pkl')
loaded_model = joblib.load('best_model.pkl')
print('Model loaded successfully:', loaded_model)
```

- The saved .pkl file must be present in your GitHub repository.

5 Bonus Tasks (Extra Points)

These tasks are optional but will earn significant extra marks. Complete as many as apply to your assigned dataset — they are what separate a good submission from a great one.

Bonus Task	Applies To	What to Do	Pts
Streamlit App	All datasets	Build a simple Streamlit web app where a user can input client/customer/student details and get a live prediction. Include app.py in your GitHub repo.	+15
F1-Score Evaluation	Classification only	Calculate and report the F1-Score for all models. Discuss which model achieves the best balance between precision and recall.	+10
ROC-AUC Score	Classification only	Plot the ROC curve for all models on a single chart. Report and compare AUC scores — explain what the curve tells you about each model.	+10
Residual & Error Analysis	Regression only	Plot residuals vs predicted values and the residual distribution. Discuss any patterns (heteroscedasticity, bias) you observe.	+10
Hyperparameter Tuning	All datasets	Apply GridSearchCV or RandomizedSearchCV on your best model. Show a before vs after performance comparison table.	+8
Feature Selection	All datasets	Apply at least ONE feature selection method — Chi-Square, ANOVA F-test, or Tree-based Feature Importance. Show selected vs dropped features.	+8
Model Comparison Table	All datasets	Create a single table comparing ALL models across ALL metrics evaluated. Make it clear which model wins on each metric.	+5
Cross-Validation	All datasets	Use 5-fold (Stratified for classification) Cross-Validation on your best model. Report the mean and standard deviation of the score across folds.	+4
Suppress Warnings & Clean Code	All datasets	Use the warnings library to suppress all warnings in your notebook. Ensure the notebook runs clean end-to-end with no red error/warning output.	+3
SHAP Explainability	All datasets	Use SHAP to explain your best model's predictions. Include a summary plot and one force plot for a sample prediction.	+3

Note: Bonus points are additive — every bonus task completed adds to your total score. Only classification-specific or regression-specific tasks relevant to your assigned dataset will be scored.

6 Scoring Rubric

Total marks: 70. Bonus marks (up to 40) are added on top.

Task	Criteria	Marks
Library Import & Setup	All mandatory libraries imported in first cell, dataset loaded and printed correctly	5
EDA	Depth of analysis, visualisation quality, insights extracted and reported	20
Feature Engineering	Encoding done correctly, scaling applied, train/test split done correctly for the problem type	15
Model Training	All 3 model categories trained for the assigned problem type, algorithm choice justified with reasoning	15
Evaluation	All required metrics reported for every model (classification or regression), visualised clearly, best model justified	10
Joblib — Model Saved	.pkl file saved, loaded back and verified, present in GitHub repo	5
Total (Must Tasks)		70
Bonus (max +40)	Top bonus tasks scored and added on top of 70	+40



7 Submission Instructions

GitHub Repository Structure

Name your repository: **MLCOE_Wildcard_YourName**

File	Description	Required?
notebook.ipynb	Complete Jupyter Notebook	MUST
best_model.pkl	Saved model using joblib	MUST
dataset file	The dataset assigned to you (bank.csv / telco_churn.csv / student_performance.csv)	MUST
README.md	Brief description: dataset used, approach, models used, best model, results summary	MUST
app.py	Streamlit app (if bonus attempted)	Bonus
requirements.txt	List of all libraries used with versions	Bonus

How to Submit

- Submit your GitHub repository link via the Google Form shared by MLCOE.
- Make your repository PUBLIC before submitting.
- Repository name format: MLCOE_Wildcard_YourName
- Clearly mention in your README which of the three datasets you worked on.

Good luck! Show us what you've got.

MLCOE is looking for passionate learners — not just coders.